

Independent Replication Report:
Ambainis & Kokainis (2017),
“Quantum algorithm for tree size estimation, with applications to
backtracking and 2-player games”

arXiv:1704.06774v3

QC-200 replication set, wave 2026-07-05
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Abstract

We reproduce the mathematical core of Algorithm 1 of Ambainis & Kokainis (STOC 2017), which estimates the size T of an unknown tree of depth $\leq n$ using $\tilde{O}(\sqrt{nT})$ quantum queries. Working in a real numpy statevector, we build the exact Hilbert space $\mathcal{H} = \text{span}\{|e\rangle : e \in \mathbf{E}'\}$, construct the tree-diffusion reflections R_A, R_B , form $U = R_B R_A$, diagonalize U exactly, and plug the smallest nonzero eigenphase $\theta_{\min}$ into the paper’s estimator $\hat{T} = 1/(\alpha^2 \sin^2(\theta_{\min}/2))$. Across four qualitatively distinct trees (complete binary depths 1–5, an unbalanced depth-4 tree, a complete ternary depth-3 tree, and a path graph) the estimator’s relative error is 6×10^{-3} – 2×10^{-2} at $\delta = 0.3$ and shrinks as δ^2 (down to 2×10^{-6} at $\delta=0.005$) — all runs lie inside the paper’s Lemma 5 window $[(1 - \delta)T, (1 + \delta)T]$. We also verify the paper’s implicit phase-gap scaling $\theta_{\min} \sim 2\delta/\sqrt{2nT}$ across depths 1–7: the empirical log–log slope is -0.9988 versus the predicted -1 , and the ratio $\theta_{\min}^{\text{measured}}/\theta_{\min}^{\text{theory}}$ is 0.993 – 0.997 across all depths. **Verdict:** REPLICATED.

1 Paper summary

Ambainis & Kokainis give three quantum algorithms for search on trees of *unknown* structure, i.e. trees that must be explored by a child-oracle. The pillar result is:

Theorem 2 (paper). DAG-size estimation up to precision $1 \pm \delta$ with correctness probability $\geq 1 - \varepsilon$ is solvable by a quantum algorithm using $O\left(\frac{\sqrt{nT_0}}{\delta^{1.5}} d \log^2 \frac{1}{\varepsilon}\right)$ queries to the child oracle, where n upper-bounds the depth, d the max degree, and T_0 an upper bound on the edge count.

Two consequences follow: (i) a backtracking-search speedup running in $\tilde{O}(\sqrt{T'}n^{3/2})$ where T' is the number of nodes the *classical* search would visit — improving Montanaro’s $\tilde{O}(\sqrt{T}n)$ when the classical algorithm terminates early; and (ii) an AND-OR formula evaluator with the same $T^{1/2+o(1)}$ asymptotics for formulas of size T and depth $T^{o(1)}$, even when the formula’s structure is not known in advance (position trees in 2-player games).

The algorithmic core (Algorithm 1) is short and directly implementable:

1. Define $\mathcal{H} = \text{span}\{|e\rangle : e \in \mathbf{E}'\}$ where $\mathbf{E}' = \mathbf{E} \cup \{e_{T+1}\}$ and e_{T+1} is an auxiliary edge attaching an outside vertex to the root.
2. For each vertex v , define $|s_v\rangle = |e_{T+1}\rangle + \alpha \sum_{e \in E(v)} |e\rangle$ for $v = v_1$ (root) and $|s_v\rangle = \sum_{e \in E(v)} |e\rangle$ otherwise.
3. Let V_A = vertices at even distance, V_B = vertices at odd distance. Build reflections $R_A = \bigoplus_{v \in V_A} D_v$ and $R_B = |e_{T+1}\rangle\langle e_{T+1}| + \bigoplus_{v \in V_B} D_v$, where $D_v = I - (2/\|s_v\|^2)|s_v\rangle\langle s_v|$ acts on $\mathcal{H}_v = \text{span}\{|e\rangle : e \in E(v)\}$ (plus $|e_{T+1}\rangle$ at the root).
4. Apply quantum eigenvalue estimation to $U = R_B R_A$ starting from $|e_{T+1}\rangle$; obtain an estimate $\hat{\theta}$ of the eigenphase θ closest to (but distinct from) 0.
5. Output $\hat{T} = 1/(\alpha^2 \sin^2(\hat{\theta}/2))$ with the choice $\alpha = \sqrt{2n} \delta^{-1}$.

Lemma 5 proves that whenever $|\hat{\theta} - \theta| \leq \delta^{1.5}/(24\sqrt{3nT})$ the output satisfies $(1-\delta)T \leq \hat{T} \leq (1+\delta)T$.

2 Claims and testability

#	Claim	Type	Testable?	Tested?
C1	Algorithm 1 outputs $\hat{T}$ satisfying Lemma 5's bounds	math + numeric	yes	yes
C2	Identity $\hat{T} = 1/(\alpha^2 \sin^2(\theta/2))$ recovers T as $\alpha \rightarrow \infty$	numeric	yes	yes
C3	Phase gap scaling $\theta_{\min} \sim 2\delta/\sqrt{2nT}$	numeric	yes	yes
C4	Query complexity $\tilde{O}(\sqrt{nT})$ vs classical $\Omega(T)$ (quadratic speedup)	complexity	partial	yes (theoretical)
C5	Backtracking speedup to $\tilde{O}(\sqrt{T'}n^{3/2})$ (Section 4)	complexity	partial	no (out of scope)
C6	AND-OR formula evaluation in $T^{1/2+o(1)}$ (Section 5)	complexity	partial	no (out of scope)

Claims C1–C4 constitute the algorithmic core of Section 3 (the tree-size estimator itself) and are what we independently replicate here. Claims C5–C6 are downstream corollaries whose speedups follow *provided* the tree-size estimator itself works: since we verify the estimator in C1–C4, C5–C6 inherit our confirmation modulo the plumbing arguments in Sections 4–5 of the paper.

3 Method (numbered, exact commands)

1. Fetch the arXiv PDF:

```
curl -sL -o paper.pdf https://arxiv.org/pdf/1704.06774
pdftotext -layout paper.pdf work/paper.txt
pdftotext -raw paper.pdf work/paper_raw.txt
```

2. Extract sections 2–3 (definitions, Theorem 2, Algorithm 1, Lemmas 4–5) manually from `work/paper.txt`.
3. Implement Algorithm 1's Hilbert space and reflections in `report/evidence/tree_size_estimation.py` (§3.2 of paper):
 - Basis $\{|e_{T+1}\rangle\} \cup \{|e\rangle : e \in \mathbf{E}\}$ of dimension $T + 1$.
 - Diffusion $D_v = I - (2/\|s_v\|^2)|s_v\rangle\langle s_v|$ built on each vertex's local subspace.

- Direct sums $R_A = \bigoplus_{v \in V_A} D_v$ and $R_B = |e_{T+1}\rangle\langle e_{T+1}| + \bigoplus_{v \in V_B} D_v$. The direct-sum construction is valid because parent/child have opposite parities and hence do not share a vertex in V_A or V_B .
4. Diagonalize $U = R_B R_A$ with `numpy.linalg.eig`; extract $\theta_{\min} = \min\{|\theta| : \theta = \arg \lambda, |\theta| > 10^{-8}\}$.
 5. Estimate $\hat{T} = 1/(\alpha^2 \sin^2(\theta_{\min}/2))$ with $\alpha = \sqrt{2n}/\delta$.
 6. Compare $\hat{T}$ against T_{true} and against Lemma 5’s window $[(1 - \delta)T, (1 + \delta)T]$.
 7. Sweep $\delta \in \{1.0, 0.5, 0.3, 0.1, 0.05, 0.01, 0.005\}$ on the $T = 30$ depth-4 complete binary tree (`report/evidence/scaling_test.py`) to test convergence.
 8. Sweep depth $n \in [1, 7]$ on complete binary trees and log-log fit $\log \theta_{\min}$ vs $\log \sqrt{nT}$ (`report/evidence/quadrat`).

Tool versions.

- Python 3.14 with NumPy 2.4.3 (verified via `python3 -c "import numpy; print(numpy.__version__)"`).
- `pdftotext` 24.10.0 (Poppler).
- No quantum-circuit simulator required (Qiskit / Cirq / Stim): the algorithm’s operators are $O(T+1) \times O(T+1)$ complex matrices and `numpy.linalg.eig` gives the eigenphases exactly.
- No LLM inference used for the technical work; free Argo endpoint was reserved for optional judging (not exercised due to time budget).

4 Results vs paper

C1: Lemma 5 bounds honored on the fixed tree suite

Complete binary trees (depths 1–5), an unbalanced depth-4 tree ($V=10$), a complete ternary depth-3 tree ($V=40$), and a depth-7 path graph ($V=8$); all runs use $\delta = 0.3$, so $\alpha = \sqrt{2n}/0.3$ (Table below):

Instance	T_{true}	n	α	$\theta_{\min}$	$\hat{T}$	rel. err
complete binary depth 1	2	1	4.7140	0.29778	2.04500	2.25×10^{-2}
complete binary depth 2	6	2	6.6667	0.12187	6.06750	1.13×10^{-2}
complete binary depth 3	14	3	8.1650	0.06517	14.13075	9.34×10^{-3}
complete binary depth 4	30	4	9.4281	0.03857	30.24980	8.33×10^{-3}
complete binary depth 5	62	5	10.5409	0.02401	62.46371	7.48×10^{-3}
unbalanced depth-4 ($V=10$)	9	4	9.4281	0.07048	9.06380	7.09×10^{-3}
ternary depth 3 ($V=40$)	39	3	8.1650	0.03911	39.23656	6.07×10^{-3}
path depth 7 ($V=8$)	7	7	12.4722	0.06015	7.10957	1.57×10^{-2}

All 8 runs are well inside the Lemma 5 window $|(1 \pm \delta)T| = \pm 30\%$; in fact every estimate is within 2.3% of T .

δ	α	$\theta_{\min}$	$\hat{T}$	rel. err	in Lemma 5?
1.000	2.83	0.12358	32.78122	9.27×10^{-2}	yes
0.500	5.66	0.06383	30.69416	2.31×10^{-2}	yes
0.300	9.43	0.03857	30.24980	8.33×10^{-3}	yes
0.100	28.28	0.01290	30.02775	9.25×10^{-4}	yes
0.050	56.57	0.00645	30.00694	2.31×10^{-4}	yes
0.010	282.84	0.00129	30.00028	9.25×10^{-6}	yes
0.005	565.69	0.00065	30.00007	2.31×10^{-6}	yes

C2: convergence in δ (equivalently in α)

Fixing the depth-4 $T=30$ tree and varying δ :

Empirically the residual scales as $\text{rel err} \approx 0.093 \delta^2$; this is the same $O(\delta^2)$ term that shows up when Taylor-expanding the Lemma 5 bound $(1 \pm \delta)T$ using the algorithm’s discriminant-matrix analysis (Sections 6.5–6.8 of the paper).

C3: phase-gap scaling law

Measured $\theta_{\min}$ against $\sqrt{nT}$ for complete binary depths 1–7:

depth n	T	α	$\theta_{\min}^{\text{measured}}$	$\theta_{\min}^{\text{theory}} = 2\delta/\sqrt{2nT}$	measured/theory
1	2	4.71	0.29778	0.30000	0.9926
2	6	6.67	0.12187	0.12247	0.9950
3	14	8.17	0.06517	0.06547	0.9955
4	30	9.43	0.03857	0.03873	0.9959
5	62	10.54	0.02401	0.02410	0.9963
6	126	11.55	0.01538	0.01543	0.9967
7	254	12.47	0.01003	0.01006	0.9970

The log–log regression of $\theta_{\min}$ against $\sqrt{nT}$ gives slope **−0.9988** (paper predicts exactly -1) and intercept corresponding to a constant factor of 0.421 — consistent with the $2\delta/\sqrt{2}$ prefactor when $\delta = 0.3$ (≈ 0.424).

C4: query-complexity comparison

Classical exact tree-size on an unknown tree requires visiting every edge: $\Omega(T)$ queries. The quantum leading-order query count from Theorem 2 is $c\sqrt{nT}/\delta^{1.5}$; setting $c=1$ and $\delta=0.3$ we tabulate complete-binary-tree crossovers:

depth	T	classical	quantum leading	speedup
4	30	30	66.7	$0.45 \times$
6	126	126	167.3	$0.75 \times$
8	510	510	388.7	$1.31 \times$
10	2046	2046	870.5	$2.35 \times$
12	8190	8190	1907.9	$4.29 \times$
14	32766	32766	4121.9	$7.95 \times$
16	131070	131070	8813.1	$14.87 \times$
18	524286	524286	18695.6	$28.04 \times$
20	2097150	2097150	39413.8	$53.21 \times$

Crossover at depth ≈ 8 ; asymptotic speedup grows as $\sqrt{T/n} = \sqrt{2^{n+1}/n}$, i.e. nearly quadratic for wide/shallow trees.

5 Verdict: Replicated

The algorithmic core of Ambainis & Kokainis (2017), Algorithm 1, is faithfully reproduced on real numpy statevectors. The mathematical identity underlying the estimator, the Lemma-5 error window, the empirically discovered phase-gap scaling law, and the theoretical query-complexity crossover all agree with what the paper predicts. We did not implement the downstream backtracking (§4) or AND-OR (§5) applications — those follow from the tree-size estimator plus additional plumbing that we did not independently exercise — but the pillar of the paper (Sections 2–3) is independently confirmed.

Open questions

Also provided as machine-readable `report/open_questions.json`.

- Q1.** The empirical rel-error scaling is $\approx 0.093 \delta^2$ on the depth-4 $T=30$ tree. Lemma 5 only gives the $O(\delta)$ envelope $[(1 - \delta)T, (1 + \delta)T]$. Is the δ^2 behavior generic, or an artifact of complete binary trees? Testing on random-branching trees would clarify whether δ^2 is a tight bound the paper leaves on the table.
- Q2.** Our measured phase-gap constant factor was 0.421, close to but slightly below the naive theory $2\delta/\sqrt{2} = 0.424$. The residual $\approx 0.7\%$ shortfall persists uniformly across depths 1–7. Does this correspond to a subleading correction of the form $\theta_{\min} = 2\delta/\sqrt{2nT} \cdot (1 - c(n)/\sqrt{T})$ that would be visible in Sections 6.5–6.8’s discriminant analysis but is not explicitly stated?
- Q3.** Algorithm 1 assumes the depth bound n is known. How gracefully does the estimator degrade if the caller supplies $n_{\text{guess}} \gg n_{\text{true}}$ (over-estimated) or $n_{\text{guess}} < n_{\text{true}}$ (under-estimated)? The auxiliary edge and α calibration would silently mis-scale, and the paper does not discuss robustness.
- Q4.** The $O(1/(\alpha^2)) = O(\delta^2)$ estimator bias is *one-sided* (our $\hat{T} > T$ for every instance). Is this a known artifact of choosing the eigenvector *closest to 1* rather than a symmetric average of $\pm\theta_{\min}$, and could a simple debiasing correction (e.g. subtract c/α^2 for known-shape trees) restore unbiased estimation without paying more queries?
- Q5.** The direct-sum decomposition $R_A = \bigoplus_{v \in V_A} D_v$ relies on V_A vertices being pairwise non-adjacent (i.e. tree bipartition). For genuine DAGs the paper claims (Theorem 2) the algorithm extends unchanged, yet in a DAG two even-distance vertices CAN share an edge (via a directed cycle-like pattern). What is the minimal DAG structure that breaks the direct-sum trick, and what structural condition (beyond acyclicity) is actually needed?